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Palaeomagnetic study of Llandovery (Lower Silurian) red beds in north-west England

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SUMMARY

High unblocking temperature magnetization components carried by haematite have been isolated from 12 sites of Llandovery (Lower Silurian) red mudstones from NW England. The principal deformation phase in this area is Acadian (mid-Devonian), however, some faults (with related folds) were also active during the later Variscan orogeny. The site-mean magnetization directions pass a regional fold test at the 95 per cent confidence level, implying that the magnetization components pre-date Acadian folding. The overall mean direction (Declination: 43.4°, Inclination: −24.1°, α_{95}: 12.4°) gives a pole position at 13.6°S, 313.9°E and a palaeolatitude of 12°S. The low palaeolatitude implies closure of the Iapetus Ocean between Eastern Avalonia and Laurentia by Late Llandovery time and is consistent with a recently published palaeolatitude (13°S) for the Wenlock (mid-Silurian) of southern Britain. Comparison of these values with the published palaeolatitude from NW England of 43°S for Caradoc (Late Ordovician) time implies very rapid (at least 15 cm yr^{-1}) northward drift of Eastern Avalonia during Late Ordovician/Early Silurian time. The new pole position from the Llandovery red beds lies close to Late Silurian and Early Devonian poles from Scotland and Wales, but is significantly different from a group of mid-Silurian poles comprising data from Scottish intrusions and from Swedish sediments. This discrepancy may be due to Late Silurian/Early Devonian (pre-Acadian) remagnetization of our sampling sites, or local clockwise rotation of the sampling sites during Acadian and/or Variscan deformation. However, the positive fold test, presence of reversals and local deformation history tend to mitigate against these possibilities. The more likely interpretation is that although Baltica/Scotland were one continental block from the mid-Silurian (Scandian orogeny), Eastern Avalonia did not become part of this block until latest Silurian.

Key words: England, Iapetus, Llandovery, palaeomagnetism.

1 INTRODUCTION

Recent Paleozoic palaeomagnetic data from Britain south of the Iapetus suture (Eastern Avalonia) have demonstrated that a wide (at least 3500 km) Arenig–Llanvirn (Early to mid-Ordovician) Iapetus Ocean between Eastern Avalonia and Laurentia closed rapidly during Late Ordovician and Silurian time. The northward motion of Eastern Avalonia has been documented by estimates of palaeolatitude which decrease from 62°S for the Arenig (Trench *et al.* 1992), to 54°S for the Llanvirn (McCabe & Channell 1990; McCabe, Channell & Woodcock 1992), to 43°S for Caradoc (Channell & McCabe 1992) and 17°S for the Early Devonian (Channell, McCabe & Woodcock 1992a). These new results are from Wales (see review by Channell *et al.* 1992b) apart from the Caradoc result which is from the English Lake District (NW England). An additional new result from Wenlock (mid-Silurian) lavas from southern Britain has yielded a palaeolatitude estimate of 13°S (Torsvik *et al.* 1993). These palaeolatitudes for Eastern Avalonia imply very rapid closure of the Iapetus Ocean between Eastern Avalonia and Laurentia during Late Ordovician and Silurian time. The new Wenlock result implies that this part of the Iapetus Ocean was more or less closed by Wenlock time,

consistent with sediment source arguments (Soper & Woodcock 1990). Palaeomagnetic data from Baltica illustrate a similar northward motion in Late Ordovician time (Torsvik *et al.* 1992), however, detailed comparison of polar wander paths for Baltica and Eastern Avalonia is inhibited by tectonic rotations which have clearly affected some of the data (viz. the Arenig and Wenlock results from Eastern Avalonia). Such rotations do not, of course, affect the palaeolatitude estimates, but the polar wander path for Eastern Avalonia remains incomplete and cannot be unambiguously compared with that for Baltica.

A principal void in the Eastern Avalonian apparent polar wander path (APWP) is for the Silurian. The Caradoc (Late Ordovician) pole position (Channell & McCabe 1992) is far from the Early Devonian pole (Channell *et al.* 1992a) and there is no intervening result of Silurian age, other than the tectonically rotated Wenlock result (Torsvik *et al.* 1993) which now supersedes the result of Piper (1975) from the same rocks. In this paper, we report palaeomagnetic data from 12 sites of Llandovery (Lower Silurian) red beds from NW England.

2 GEOLOGICAL SETTING

The subject of this study is a unit of red marine mudstones within the Llandovery Series (Lower Silurian) of north-west England (Fig. 1). Their biostratigraphic position is tightly constrained to the *crenulata* biozone of the Telychian, the uppermost stage of the Llandovery (Fig. 2) (Rickards 1973, 1978). Lithostratigraphically the red beds occur within the upper part of the Browgill Formation (Stockdale Subgroup), within the Windermere Group (stratigraphy revised by Kneller *et al.* 1993).

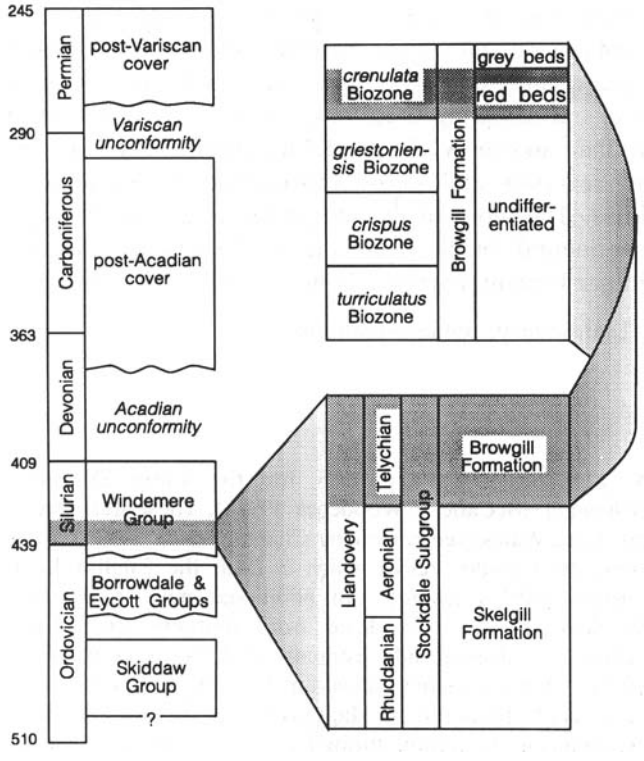


Figure 1. Stratigraphic position of the red beds within the Browgill Formation of NW England.

The oldest known exposed rocks in north-west England are the Skiddaw Group (Figs 1 and 2; Molyneux & Rushton 1984). They record marine turbiditic sedimentation on the northern margin of the Eastern Avalonia microcontinent, probably during and after the rifting of this fragment from the Gondwana continent to the south. The overlying Eycott and Borrowdale Volcanic Groups, possibly time equivalent (Millward & Molyneux 1992), preserve parts of a subduction arc built on the leading edge of Eastern Avalonia as it moved rapidly northward during Late Ordovician time. The arc, by then dominantly subaerial, shut down and subsided in mid-Caradoc to early Ashgill time. This allowed shallow marine sediments at the base of the Windermere Group to onlap the Borrowdale Group over an erosional unconformity.

Subsidence of the Windermere Group basement exceeded the supply of available sediment by Early Silurian time, resulting in graptolite-bearing mudstones and siltstones throughout Llandovery and Wenlock time (Early Silurian). The majority of these sediments are gray, with their contained iron reduced during burial. Some are black finely laminated mudstones, taken to indicate deposition as well as diagenesis in anoxic waters. By contrast, the red beds contain oxidized iron in the form of haematite. They occur in localized lenticular units up to 20 m thick and several kilometres across. Contained graptolites and trilobites confirm their deep marine origin (Rickards 1978; Hemsley 1978). This anomalous facies is sporadically present in other Llandovery sequences in the Caledonian/Appalachian orogen, and has been attributed to increased supply of detrital haematite during eustatic marine transgressions (Ziegler & McKerrow 1975).

Clastic supply to the Windermere Group increased progressively during Late Silurian time, probably in a foreland basin developed to the south of the overriding Laurentian continent (Kneller 1991). The basin was filled to sea-level by latest Silurian time and probably had an Early Devonian non-marine history. However, this uppermost part of the sequence was uplifted and totally eroded during the Acadian Orogeny, which culminated in late Early to Mid-Devonian time (review by Soper, Webb & Woodcock 1987). This deformation produced folds and cleavage trending about east–west in the sampled areas, but with a north-easterly regional trend (Soper *et al.* 1987). After deposition of a post-Acadian sedimentary cover, two of the sampled areas (Cross Fell Inlier and Howgill Fells; Fig. 2) were affected by Variscan faults and related folds, though without an associated cleavage. The interference of Variscan and Acadian faults produces the domal outcrop patterns of the Howgill Fells area (Fig. 2b). The sample sites represent all the recorded red bed localities in north-west England that are amenable to field sampling. A more detailed map of the Howgill Fells area has been published by Rickards (1967), and of the Cross Fell locality by Burgess & Wadge (1974).

3 PALAEOMAGNETIC ANALYSIS

A total of 14 sites of Llandovery red mudstones were collected using a gasoline-powered drill. 11 to 16 samples were collected at each site (Table 1). Samples were oriented using a magnetic compass. Of the 14 sites, 12 are located in

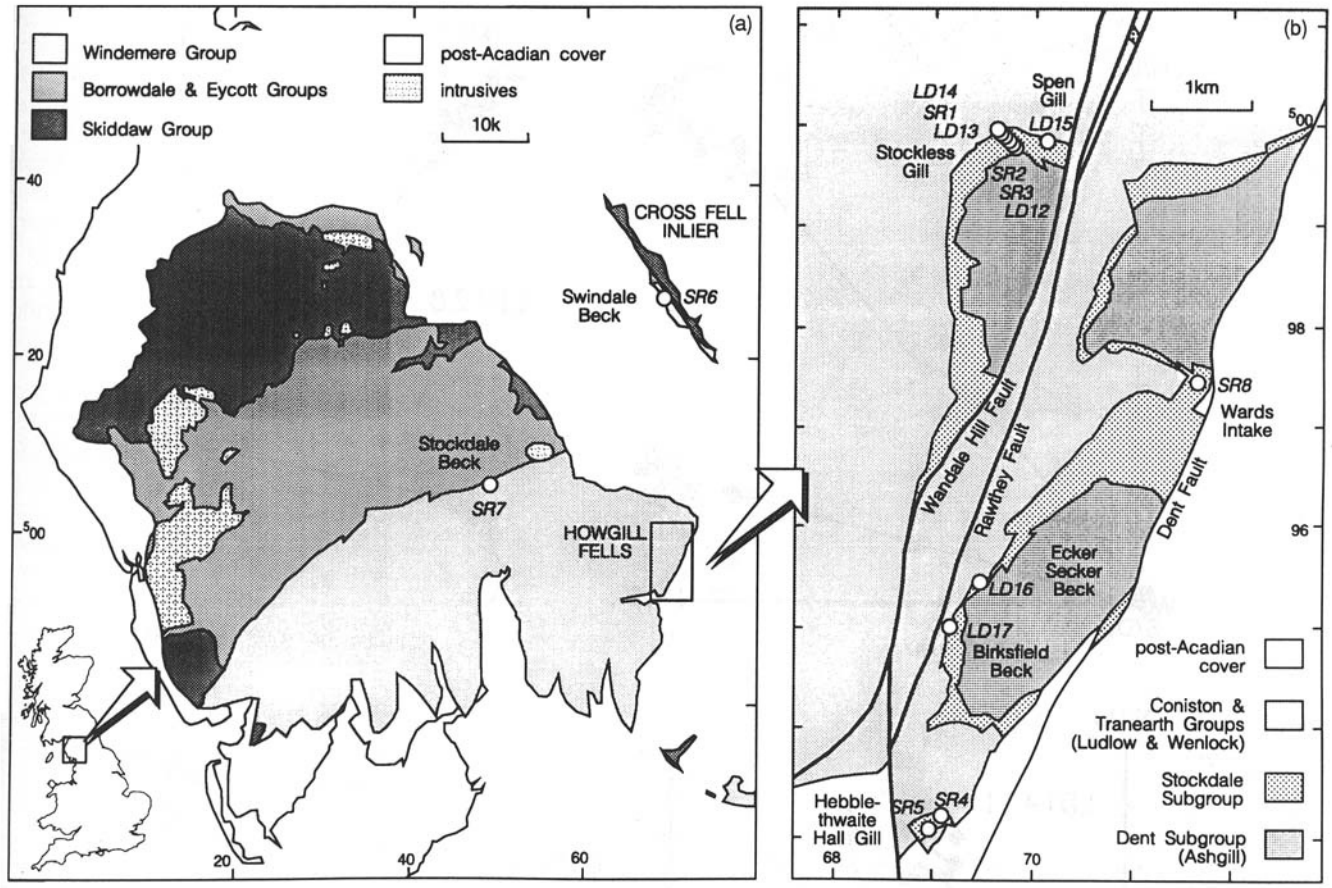


Figure 2. Location of sampling sites in NW England.

Table 1. Site-mean magnetization directions. N/No refers to the ratio of number of samples in which the high-unblocking temperature-magnetization component could be resolved, to the number of samples collected at the site. N/R gives the number samples at each site with normal- and reversed-polarity magnetization components. Grid Ref. refers to the national grid reference. Dec. and Inc. denote declination and inclination, respectively. Statistical parameters after Fisher (1953). Inclination-only analysis after Kono (1980).

				Before Tilt		After Tilt			
Site	Grid Ref.	N/No	N/R	Dec.(°)	Inc.(°)	Dec.(°)	Inc.(°)	k	α_{95}
SR1	SD 696 999	8/14	6/2	21.7	-25.6	29.9	-23.4	52.2	7.7
SR2	SD 697 999	12/13	10/2	38.7	-52.4	33.5	-31.1	253.9	2.7
SR3	SD 697 998	14/14	7/7	226.1	5.5	238.9	28.7	183.2	2.9
SR4	SD 691 932	12/15	0/12	212.8	-14.2	213.8	18.4	27.0	8.5
SR5	SD 690 931	6/12	2/4	263.0	47.4	225.4	34.8	48.9	9.7
SR6	NY 688 273	14/15	8/6	42.9	-34.7	33.2	-21.6	40.2	6.3
SR7	NY 492 055	12/12	11/1	33.9	-59.8	1.4	-19.5	539.6	1.9
LD12	SD 698 999	15/15	0/15	228.8	-19.1	225.0	7.0	298.5	2.2
LD14	SD 696 999	15/15	15/0	20.0	6.6	30.5	-36.3	77.4	4.4
LD15	SD 700 999	10/11	10/0	54.6	-1.8	57.3	-8.7	220.3	3.3
LD16	SD 694 953	14/14	14/0	33.0	-61.1	80.4	-37.6	187.7	2.9
LD17	SD 691 949	11/12	11/0	67.9	-29.5	73.0	-6.0	75.7	5.3
MEAN		12		43.5	-23.0			5.7	20.0
						43.4	-24.1	13.1	12.4
Mean of inclinations		12			-32.3			7.6	16.8
							-23.3	27.0	8.5
POLE			Latitude	Longitude		dp	dm		
			13.6°S	313.9°E		7.1°	13.1°		

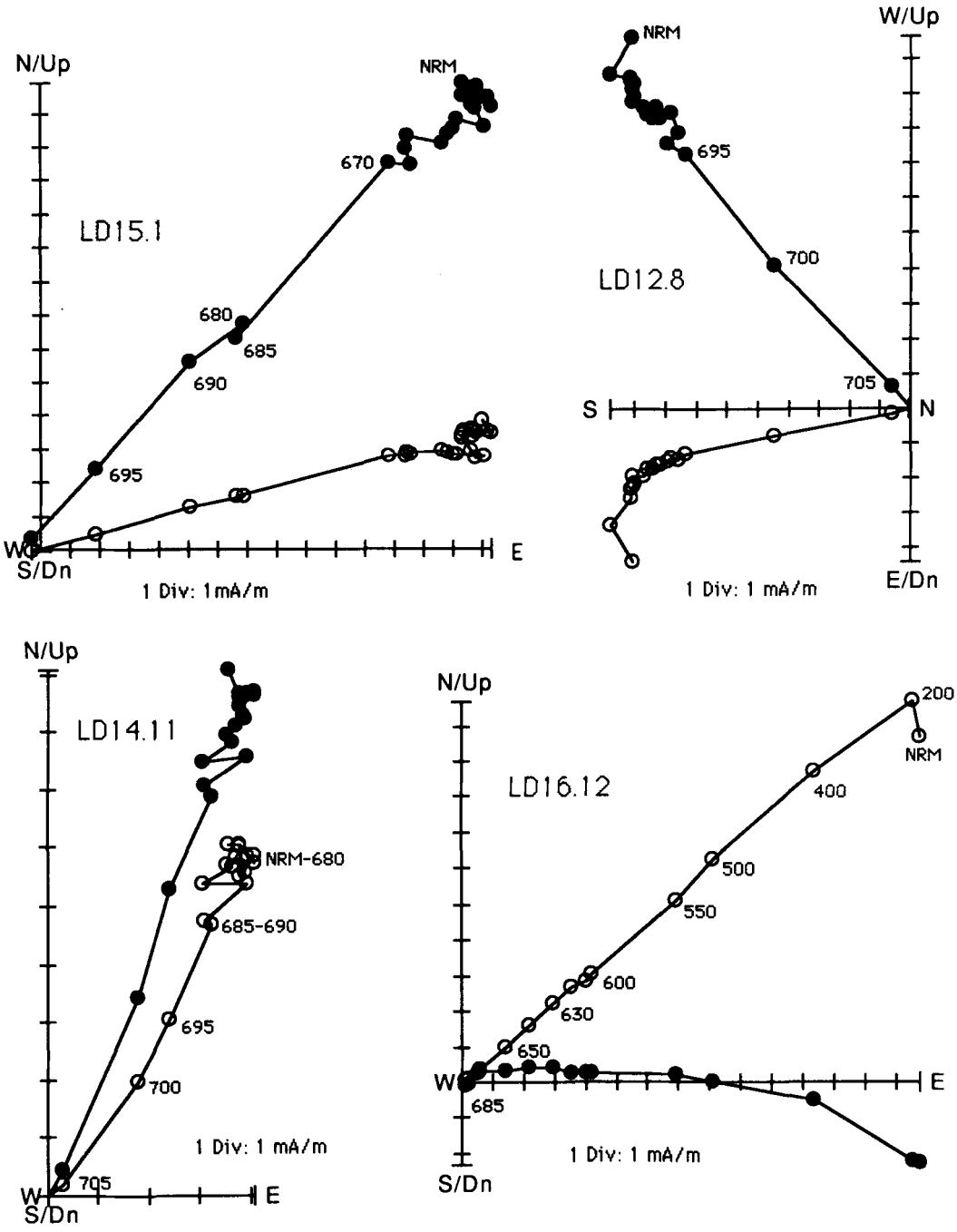


Figure 3. Orthogonal projection of thermal demagnetization data from samples showing stable demagnetization behaviour. Open and closed symbols represent projection of the vector endpoint on the vertical and horizontal plane, respectively. The demagnetization temperatures associated with some points are given in degrees Celsius.

the Howgill Fells (Fig. 2), east of the town of Sedbergh, and the other two at Stockdale Beck (near Kendal) and Swindale Beck (Cross Fell Inlier). The site distribution was dictated by the very limited outcrop of the red mudstones.

Progressive thermal demagnetization was carried out on all samples. Demagnetization temperature increments were progressively reduced from 100–200 °C at low temperatures, to 5–10 °C above 600 °C. The unblocking temperature spectra were found to be largely confined to temperatures above 600 °C (Figs 3 and 4). For some samples, the apparent maximum unblocking temperatures exceeded the Néel Temperature for haematite (Fig. 3). The unusually high-unblocking temperatures may be partly due to poor temperature calibration in the Schonstedt Thermal Demagnetizer, although some samples heated in the same demagnetizer show maximum unblocking temperatures in the predicted range (viz. LD16.12 in Fig. 3). The elevated unblocking temperatures may be partly due to defect pinning of spin moments, resulting in maximum-unblocking temperatures in excess of the Néel Temperature for pure haematite. Hysteresis parameters (Fig. 5) confirm the presence of high-coercivity remanence carriers such as

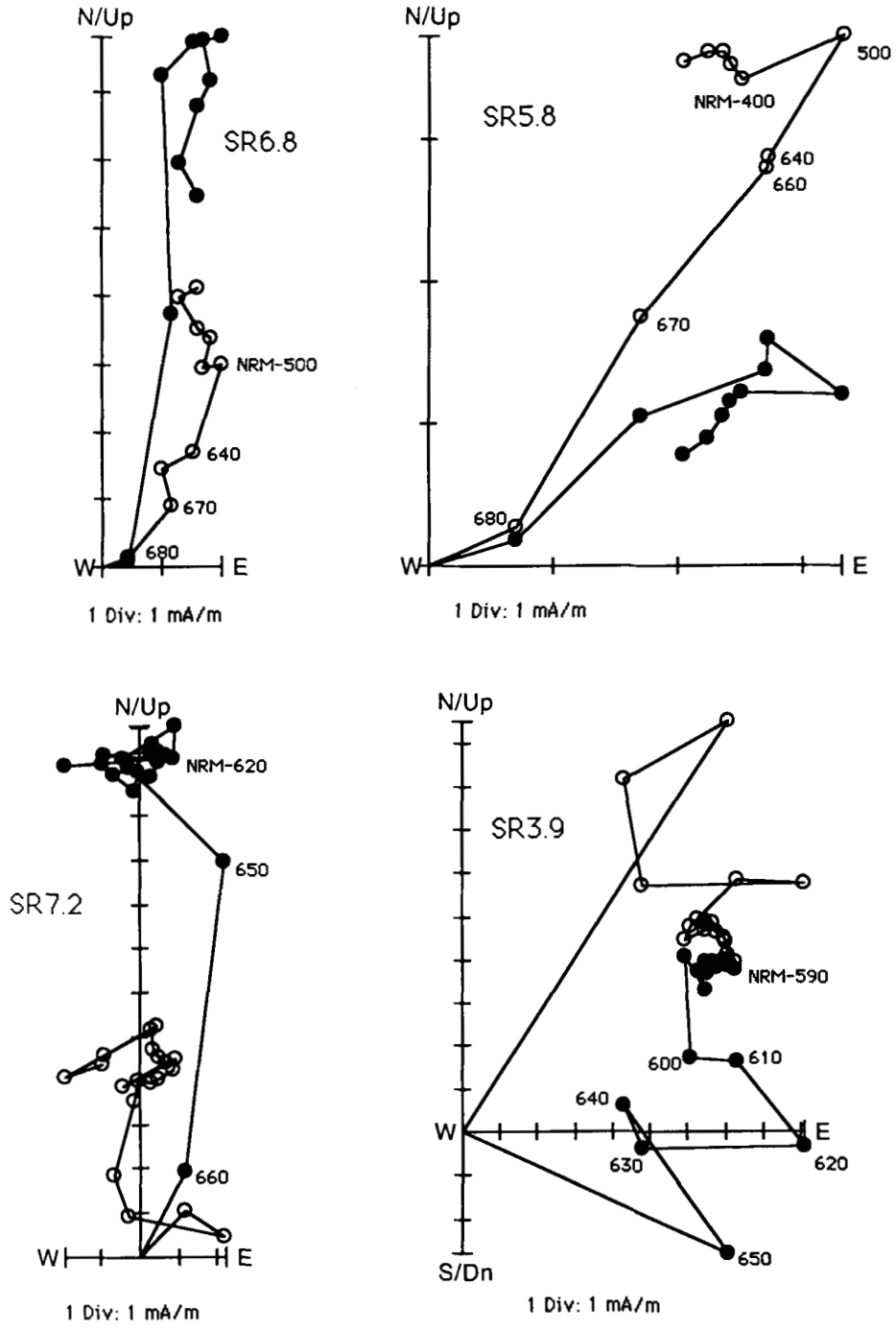


Figure 4. Orthogonal projection of thermal demagnetization data from samples showing less stable demagnetization behaviour above 600 °C. Symbols as for Fig. 3.

haematitie. For 12 of 14 sites, straight-line trajectories of vector endpoint to the origin of the orthogonal projection, indicates that a magnetization component can be resolved (Fig. 3), although for some samples the component direction is poorly defined due to erratic behaviour of the magnetization vector above 600 °C (Fig. 4). The more erratic behaviour was apparent in about 30 per cent of the samples. For some of these samples, we have computed the direction from the stable endpoint, assuming that the stable endpoint denotes a single component. The high precision of site-mean directions (Table 1) and the observation of univectorial decay of remanence in the majority of samples (Figs 3 and 4) indicates that the use of stable endpoints does not perturb the site-mean directions. Magnetization components were not resolved at two sites (LD13 and SR8) due to unstable demagnetization behaviour.

The directions of the magnetization component in individual samples were computed using the standard least-squares method (Kirschvink 1980). The component directions within each site were then combined using Fisher

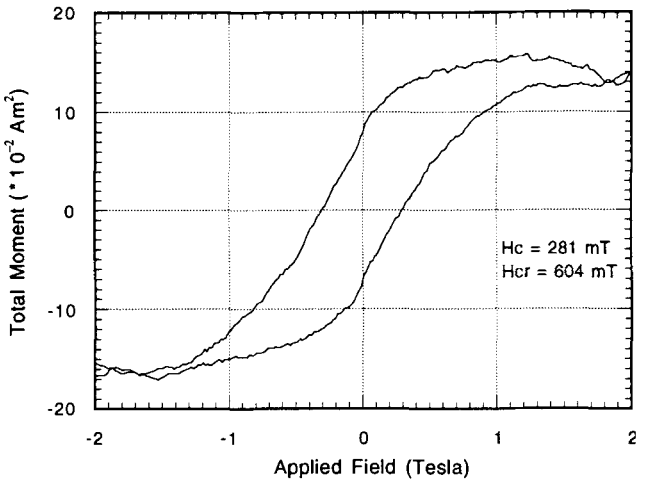


Figure 5. Hysteresis loop for a sample of red mudstone from site SR1. Hc and Hcr denote coercive force and coercivity of remanence, respectively.

(1953) statistics (Table 1). Although the within-site precision is high (Table 1), the site-mean directions show considerable scatter (Fig. 6), particularly in declination. Nonetheless, the regional fold test is positive at the 95 per cent confidence level using the criteria of McElhinny (1964) and at 99 per cent using those of McFadden (1990), implying that the magnetization components pre-date the Acadian (mid-Devonian) folding. In addition, both normal and reversed polarities were observed at several sites (Table 1), and the reversal test is positive at the 99 per cent confidence level. The scatter in site-mean declinations may be partly due to local rotation associated with Acadian and Variscan motion on the three major NE–SE trending faults which cut through the area. Two sites (LD16 & LD17) have higher declinations (Fig. 6) and this may be due to clockwise rotation associated with the nearby Rawthey Fault (Fig. 2). One isolated site (SR7) gives a site-mean direction which is rotated anticlockwise from the mean. An analysis of the inclination data, using the method of Kono (1980), shows that the precision (k) of the mean inclination, improves from 7.6 to 27.0 on tilt correction (Table 1).

4 APPARENT POLAR WANDER PATHS AND PALAEOGEOGRAPHY

The inclination-only analysis (Kono 1980) yields a mean inclination of 23.3° which passes the fold test at the 99 per cent confidence level (Table 1). This pre-Acadian inclination value is interpreted to signify a Llandovery palaeolatitude of 12.2°S (±4°) for NW England. This is consistent with an estimate of 13°S for the Wenlock of southern England (Torsvik, Løvlie & Storevedt 1993). A comparable palaeolatitude (17°S) was resolved from the Early Devonian Old Red Sandstone of south Wales (Channell *et al.* 1992a). Similarly low palaeolatitudes from Laurentia (Scotland and North America) and Baltica for Silurian and Early Devonian time imply that all three continents were in close juxtaposition at this time (see review by Channell *et al.* 1992b).

Efforts to construct Palaeozoic APWPs for Scotland (Laurentia), Baltica and Eastern Avalonia are beset by a paucity of reliable data, and by local tectonic rotations which appear to have affected some of the available data, particularly those from Eastern Avalonia. Local tectonic rotations do not, of course, affect the estimates of palaeolatitude but inhibit the use of these data for

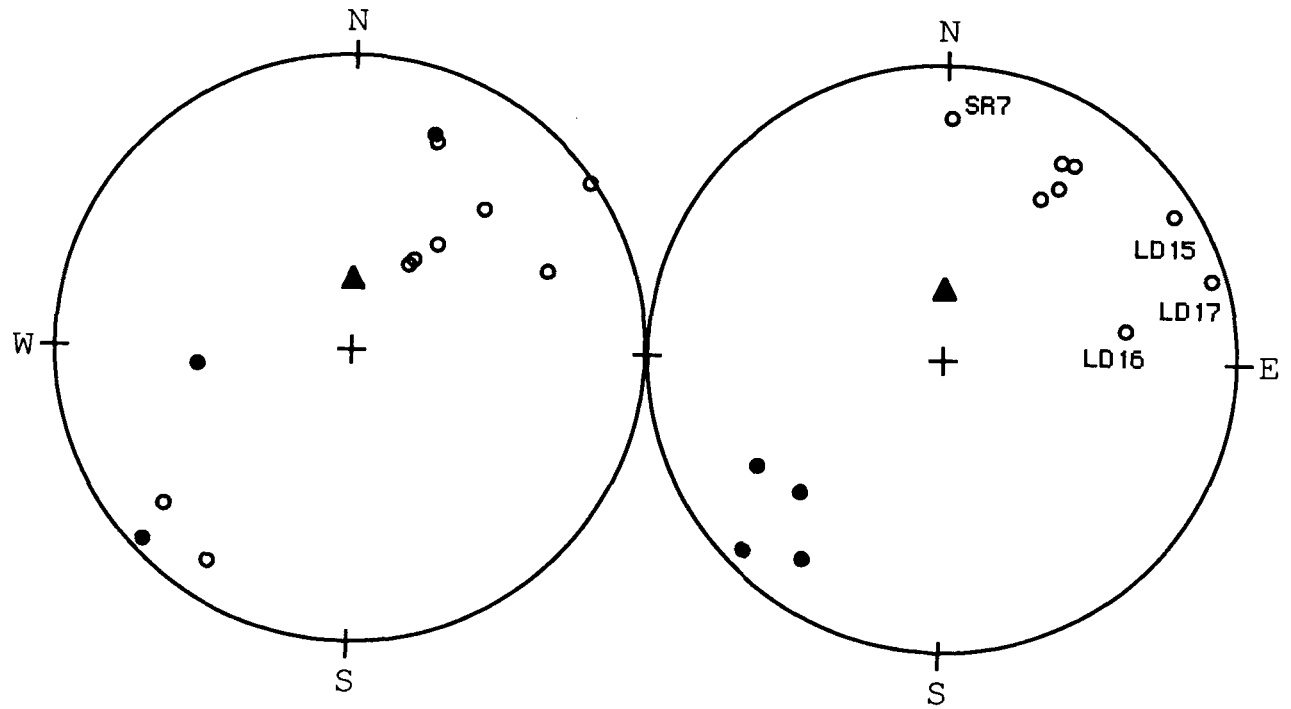


Figure 6. Equal-area projection of site-mean magnetization directions before (left) and after (right) structural tilt correction. Closed and open symbols denote downward and upward inclinations, respectively. The triangle gives the present Earth's field direction in the sampling region.

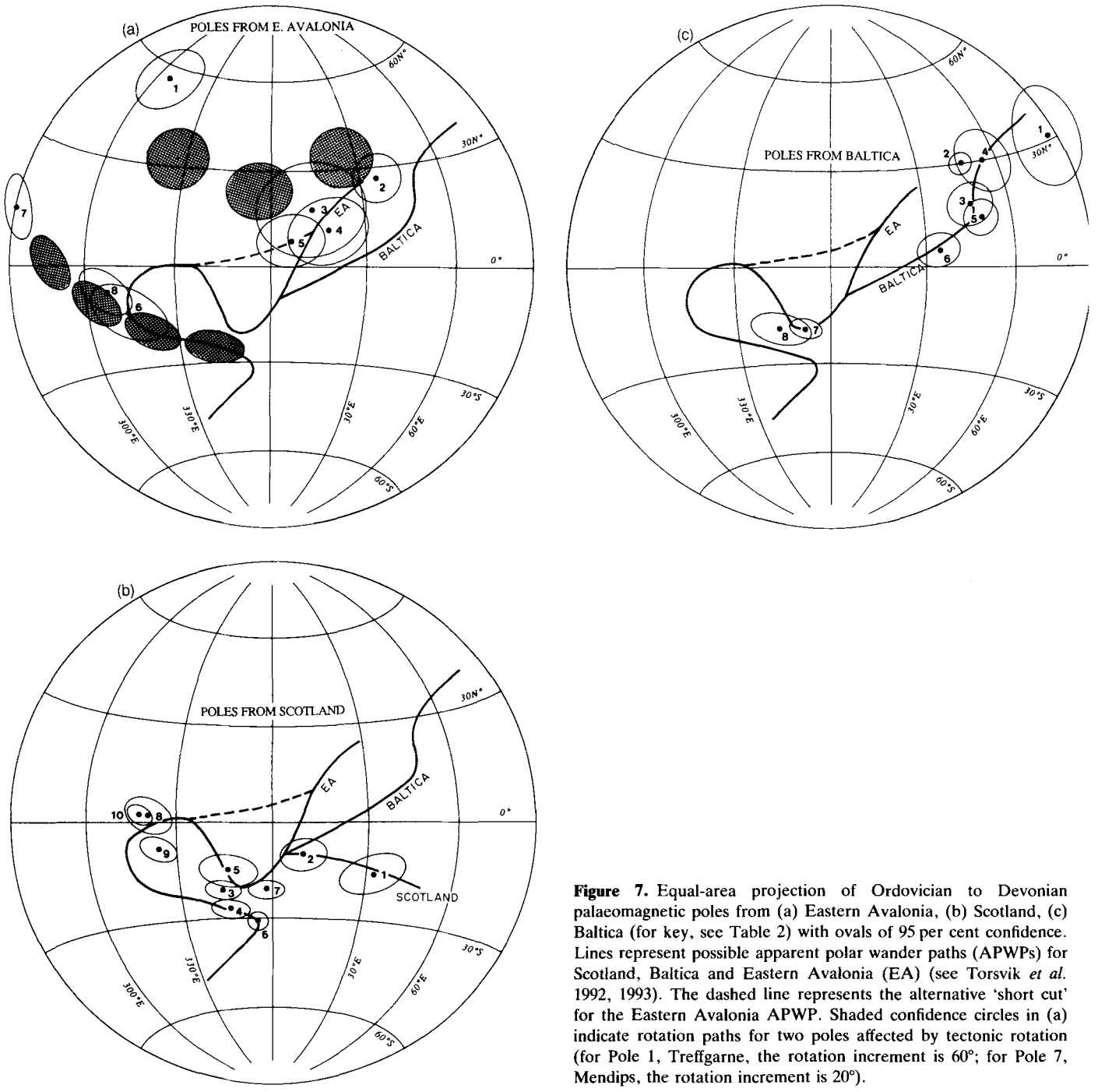


Figure 7. Equal-area projection of Ordovician to Devonian palaeomagnetic poles from (a) Eastern Avalonia, (b) Scotland, (c) Baltica (for key, see Table 2) with ovals of 95 per cent confidence. Lines represent possible apparent polar wander paths (APWPs) for Scotland, Baltica and Eastern Avalonia (EA) (see Torsvik *et al.* 1992, 1993). The dashed line represents the alternative 'short cut' for the Eastern Avalonia APWP. Shaded confidence circles in (a) indicate rotation paths for two poles affected by tectonic rotation (for Pole 1, Treffgarne, the rotation increment is 60°; for Pole 7, Mendips, the rotation increment is 20°).

construction of APWPs. The modern Ordovician to Devonian APWP for Eastern Avalonia (Fig. 7a) relies on eight poles (Table 2). Two of these poles (poles 1 and 7 in Fig. 7a) have been affected by local tectonic rotation and can be rotated back on the APWP, but do little to improve its definition (Fig. 7a).

Comparison of Scottish APWPs from the Grampian Highlands (north of the Highland Boundary Fault), Midland Valley and Southern Uplands do not indicate any significant differences, indicating that the three units have had their present juxtaposition since at least Siluro–Devonian time (Trench *et al.* 1989). Several Siluro–Devonian poles in the Scottish APWP (poles 8–10 in Fig. 7b) lie close to poles of the same age in the Eastern Avalonian APWP (poles 6 and 8 in Fig. 7a), leading to the conclusion that the Iapetus Ocean in Britain was completely closed by this time. A controversial aspect of Scottish palaeomagnetic data is the significance of the mid-Silurian 'newer' granite poles (poles 3–7 in Fig. 7b and Table 2) which plot to the SE of the Late Silurian Scottish poles and the Siluro–Devonian poles from Eastern Avalonia (Figs 7a and b). Trench *et al.* (1989) considered that the 'newer' granite poles may represent Permo–Carboniferous remagnetizations as the pole positions are located in the vicinity of Permo–Carboniferous poles. However, recently acquired mid-Silurian poles from Baltica (poles 7 and 8 in Fig. 7c) (Douglass 1988; Trench & Torsvik 1991) plot in the same position as some of the 'newer' granite poles. This has led to the acceptance of the

Table 2. Key poles which define the apparent polar wander paths for Scotland, Baltica and Eastern Avalonia in Figs 7(a)–(c).

				Pole Position		dp/dm (A_{95})
	SCOTLAND					
1	Aberdeen Gabbro	Watts & Briden, 1984	Early Ordovician.	15.6S	32.2E	5.7/10.6
2	Ledmorite	Turnell & Briden, 1983	Early. Silurian	10S	9.7E	5/7.3
3	Strontian Granite**	Torsvik, 1984	435 +/- 10 Ma	21S	344E	2.9/5.5
4	Foyers Granite**	Torsvik, 1984	400-435 Ma	27S	346E	2.9/5.8
5	Ratagan Granite**	Turnell, 1985	415 +/-5 Ma	15S	346E	5/9
6	Helmsdale Granite**	Torsvik et al. 1983	400-420 Ma	31S	355E	3.0/3.0
7	Peterhead Granite**	Torsvik, 1985	~ 415 Ma	21S	358E	2.9/5.5
8	Lorne Plateau	Latham & Briden, 1975	Late Sil/Early Dev.	2N	321E	5/8
9	Arrochar Complex	Briden, 1970	Late Sil/Early Dev.	8S	324E	3.4/6.1
10	Strathmore Lavas	Torsvik, 1985	Late Sil/Early Dev.	2.2N	318.2E	2.7/4.3
	BALTICA					
1	Norwegian Intr.	Torsvik et al. 1990	Tremadoc	31N	86E	11/15
2	Baltoscandian Lmst	Claesson, 1978	Arenig-Llanvirn	30N	46E	3/4
3	Baltoscandian Lmst	Torsvik & Trench,1991a	Arenig-Llanvirn	18N	46E	6/8
4	Baltoscandian Lmst	Perroud et al., 1992	Arenig-Llanvirn	30N	55E	(9)
5	Baltoscandian Lmst	Torsvik & Trench,1991b	Llanvirn-Llandeilo	14N	49E	5/6
6	Baltoscandian Lmst	Torsvik & Trench,1991b	Llandeilo-Caradoc	5N	34E	5/7
7	Visby Lmst.	Trench & Torsvik, 1991	Wenlock	19S	352E	3/5
8	Ringerike Sst.	Douglass, 1988	Ludlow	19S	344E	4.8/9.3
	E. AVALONIA					
1	Treffgarne*	Trench et al., 1992	Arenig	56.1N	306.5E	9/10
2	Stapeley Volcanics	McCabe & Channell,1990	Llanvirn	26.6N	36.1E	6.9/8.2
3	North Builth	McCabe et al., 1992	Llanvirn	17.9N	12.9E	14.7/17.2
4	Tramore Volc.	Deutsch, 1980	Mid Ordovician	11N	18E	10/13
5	Borrowdale Volc.	Channell & McCabe,1992	Caradoc	8.1N	6.2E	8.2/10.6
6	Red siltstones	This paper	Llandovery	13.6S	313.9E	7.1/13.1
7	Mendip Volcanics*	Torsvik et al., 1993	Wenlock	13N	271E	5/9
8	Old Red Sst.	Channell et al., 1992a	Early Devonian	7.3S	306.7E	5.4/9.5

* Poles affected by local tectonic rotation
** 'Newer' granite poles

Scottish 'newer' granite poles in recent publications and hence modification of the APWP for Scotland (Torsvik *et al.* 1992, 1993).

Torsvik *et al.* (1993) noted that their Wenlock pole from Eastern Avalonia, if rotated 80° to correct for clockwise tectonic rotation, coincides with the Ludlow and Wenlock poles from Baltica and with the poles from mid-Silurian Scottish granites (compare Figs 7a–c). This was taken as evidence that the Eastern Avalonia APWP coincides with that of Baltica/Scotland, at least from Wenlock time (Torsvik *et al.* 1993). The new Llandovery pole from Eastern Avalonia (pole 6 in Fig. 7a) does not, however, lie on the Early or mid-Silurian segment of the APWP for Baltica/Scotland. This implies that the two APWPs do not merge until the latest Silurian and that the APWP for Eastern Avalonia may take the 'short cut' shown as a dashed line in Figs 7(a)–(c). The Torsvik *et al.* (1993) Wenlock result from Britain (pole 7 in Fig. 7a) is not inconsistent with this conclusion as correction for a clockwise rotation of 50° (rather than the advocated 80°) would place this pole close to the new Llandovery pole (Fig. 7a). We suggest that the merging of the Eastern Avalonian and Baltica/Scotland APWPs was a result of the rotation of Baltica/Scotland relative to Eastern Avalonia during Wenlock and Ludlow time.

The first clear evidence in Britain and Ireland of sediment sources along the Iapetus collision zone is found in the Wenlock of southern Ireland (Murphy & Hutton 1986; Hutton & Murphy 1987). North-western sediment sources do not appear in the English Lake District (NW England) until Ludlow time, implying a diachronous closure of the Iapetus Ocean in Britain (Soper & Woodcock 1990). The final closure of the Iapetus Ocean may have involved relative rotation of Eastern Avalonia and Baltica/Scotland (Fig. 8). The merging of the APWPs from these two continental units is latest Silurian or Early Devonian in age.

5 CONCLUSIONS

The new palaeomagnetic result from the Llandovery red beds passes both a reversals test and a regional fold test at the 95 per cent or 99 per cent confidence level (depending on test criteria), suggesting that the magnetization pre-dates mid-Devonian (Acadian) folding and is probably Llandovery in age. The palaeolatitude estimate (12°S) indicates close latitudinal juxtaposition of Laurentia, Baltica and eastern Avalonia by Llandovery time and *almost* complete closure of the Iapetus Ocean in Britain (Fig. 8). The Llandovery pole position is close to the Early Devonian pole from the Old Red Sandstone of Wales (Channell *et al.* 1992a) and to Late Silurian–Early Devonian poles from Scotland (Briden 1970; Latham & Briden 1975; Torsvik 1985). The pole is, however, significantly different from a group of mid-Silurian poles comprising data from Scottish granites (Torsvik 1984; Torsvik 1985; Turnell 1985; Torsvik

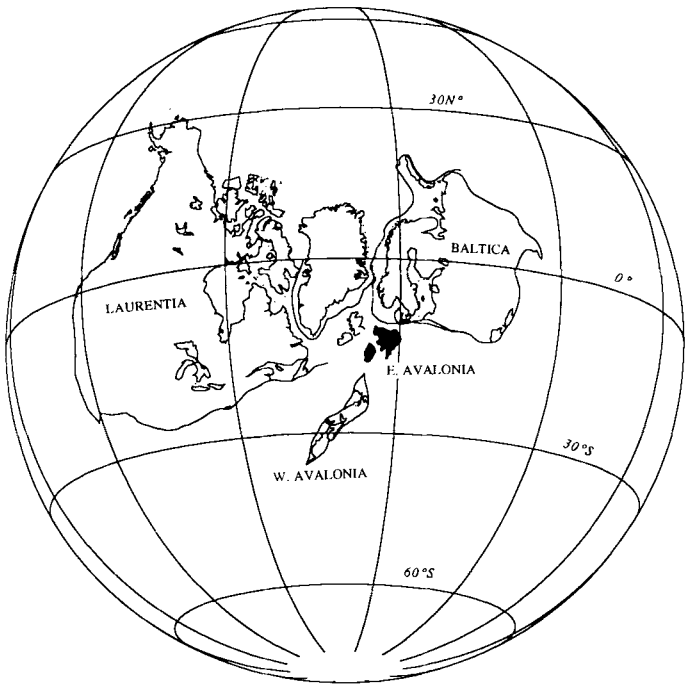


Figure 8. Late Silurian palaeogeography. The Old Red Continent (Laurentia/Scotland/Baltica) is reconstructed using the Bullard, Everett & Smith (1965) fit. Euler rotations for Europe/North America: [88.5, 27.7, Ω = 38.0], for Europe/Greenland: [73.0, 96.5, Ω = 22.0]. The Old Red Continent is positioned using the mean of seven poles comprising the 'newer' Granite poles (Scottish poles 3–7 in Table 2) and poles from the Visby Limestone and the Ringerike Sandstone (Baltica poles 7 and 8 in Table 2) [Mean Pole: Lat: 21.9°S, Long: 349.7°E, $N = 7$, $K = 124.3$, $A_{95} = 5.4$]. Eastern and Western Avalonia are positioned using the Llandovery result from this paper.

et al. 1983) and Scandanavian sediments (Douglass 1988; Trench & Torsvik 1991). At present, the pole distribution tends to indicate that although Scotland and Baltica were sutured by mid-Silurian (Scandian orogeny), Eastern Avalonia did not become part of this continental unit until Late Silurian or Early Devonian time. A clockwise rotation (of about 30°) of Eastern Avalonia relative to Scotland/Baltica in Wenlock and Ludlow time is predicted by the palaeomagnetic data.

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